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Author contributions Xiangmei Wu and Keita Ebisu conceived of the presented idea and designed the study. Nico Schulte and Scott Epstein generated the hourly air data, and Keita Ebisu processed the health outcome data. Andrew Nguyen performed data analysis, particularly designed the regionalization process with the supervision of Keita Ebisu and Xiangmei Wu. The first draft of the manuscript was written by Andrew Nguyen, Xiangmei Wu, and Keita Ebisu. The draft was reviewed and edited by Keita Ebisu, Nico Schulte, Rupa Basu, Scott Epstein, and Xiangmei Wu. All authors commented on previous versions of the manuscript and approved the final manuscript.

Data availability The hourly $PM_{2.5}$ and ozone concentration data used in this manuscript are available with a Public Records Request at. The emergency department visit data are available through the California Department of Health Care Access and Information with an approved application.

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